

Vehicle Detection Using Deep Learning: A Comparative Study of YOLO and DETR for Object Detection

Team Members

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A Summary of the Project and Its Broader Impact

Object detection is a fundamental computer vision task with applications in autonomous driving, intelligent transportation systems, traffic monitoring, and smart cities. Unlike image classification, object detection requires a model to identify both the class and location of multiple objects within an image. In this project, we will compare a convolutional neural network (CNN)-based object detector (YOLO) with a Transformer-based object detector (DETR) for vehicle detection. Using a publicly available autonomous driving dataset, we will evaluate each model's detection accuracy, computational efficiency, and inference speed. The results will provide insight into the strengths and limitations of CNN- and Transformer-based object detection methods for real-world computer vision applications.

Details of the Project

Dataset

The project will use the BDD100K, which contains thousands of annotated images collected from real-world driving environments. The dataset includes multiple object categories such as cars, trucks, buses, motorcycles, bicycles, pedestrians, and traffic signs. To keep the project focused, the primary vehicle classes used in this study will be:

- Car
- Truck
- Bus
- Motorcycle

The dataset will be divided into training, validation, and testing sets using the standard dataset split.

Models

Two object detection models will be implemented and compared:

- **YOLO** (CNN-based object detector)
- **DETR (Detection Transformer)** (Transformer-based object detector)

Both models will be initialized using publicly available pretrained weights and fine-tuned on the selected vehicle classes from the BDD100K dataset.

Training Plan

The project will be implemented using PyTorch with GPU acceleration when available and CPU support as a fallback. Images will be resized and preprocessed according to the requirements of each model. Standard data augmentation techniques, such as random horizontal flipping and color augmentation, will be applied during training. Both models will be trained using comparable training settings to ensure a fair comparison.

Evaluation Plan

Model performance will be evaluated using the following metrics:

- Mean Average Precision (mAP)
- Precision
- Recall
- F1-score
- Training time per epoch
- Inference time per image
- Number of trainable parameters

The final report will include quantitative comparison tables, precision-recall curves, sample detection visualizations, and a discussion of the trade-offs between detection accuracy, computational complexity, and inference speed.

Projection on the Achievable Outcomes

We expect both models to successfully detect common vehicle classes in driving scenes. YOLO is expected to provide faster inference and be better suited for real-time applications, while DETR may achieve competitive detection accuracy in complex scenes at the expense of increased computational cost. The project will highlight the advantages and trade-offs of CNN-based and Transformer-based object detection models and provide practical recommendations for selecting an appropriate model based on application requirements.

Individual Responsibility

David Ho

David Ho will coordinate the project timeline, prepare and preprocess the dataset, implement and train the YOLO model, evaluate model performance, maintain the GitHub repository, integrate experimental results, and prepare the final report and presentation.

Mahmoud Abdulkareem

Mahmoud Abdulkareem will implement and train the DETR model, assist with experiment execution and hyperparameter tuning, generate evaluation metrics and visualizations, compare the performance of both models, and contribute to writing, editing, and reviewing the final report.